

Digital Equipment Corporation
Maynard, Massachusetts

MR8E-C ROM Contents

[Disassembled and Reconstructed]



MR8E-C

Name: MR8E-C ROM CONTENTS
Document: MD-08-DHMRA-A-D
Kit: AC-6339A-MA
Binary P/T: MD-08-DHMRA-A-PB
Binary P/T Kit: AK-6342A-MA
File[s]: MR8EC.LST
P?S Index: O>R,LQ7D,B0037QNL0037BNMSDMSR
Security Level: 2
Location: NY
Date: 14-Feb-2016
Medium: PDF

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2	2			2
3	3	/	SOURCE RECREATION BY:	3
4	4			4
5	5	/	CHARLES LASNER P?S NY	5
6	6			6
7	7	/	LAST EDIT: 14-FEB-2016 16:00:00 CJL	7
8	17			8
9	18	/	THE MR8E-C ROM IS USED WITH THE PRIMARY TD8E CONTROLLER [DEVICE CODE 77] TO	9
10	19	/	ACHIEVE BOOTABLE OPERATING SYSTEMS REQUIRING MINIMAL READ/WRITE MEMORY. THE	10
11	20	/	PDP-8 CONVENTION OF THE ENTIRE SYSTEM DEVICE HANDLER BEING CONTAINED IN THE	11
12	21	/	READ/WRITE MEMORY OF LOCATIONS 07600-07777 CAN BE REALIZED [PERHAPS WITH SOME	12
13	22	/	EFFORT]. DEPENDING ON THE PARTICULAR OPERATING SYSTEM, THIS COULD BE A	13
14	23	/	MANDATORY REQUIREMENT; ALTERNATIVELY, THE OPERATING SYSTEM MUST BE	14
15	24	/	RECONFIGURED TO HANDLE THE SYSTEM REQUIREMENTS IN SOME OTHER SYSTEM-DEPENDENT	15
16	25	/	MANNER [SUCH AS REQUIRING ADDITIONAL MEMORY] WHICH IMPACTS ON OTHER ASPECTS OF	16
17	26	/	THE OPERATING SYSTEM AND/OR MACHINE COSTS. THE MR8E-C ROM ALLOWS A VIABLE	17
18	27	/	SOLUTION AS LONG AS THE DEVICE HANDLER CAN FIT INTO THE DESIGNATED CODE SPACE	18
19	28	/	OF 07600-07777 [WITH EACH SYSTEM RESERVING CERTAIN INTERNAL LOCATIONS THAT	19
20	29	/	INTERACT WITH THE ROM] WITHOUT UNDUE COMPROMISE.	20
21	30			21
22	31	/	SINCE PDP-8 MEMORY CONVENTIONS ARE SUCH THAT ONLY ENTIRE 4096 WORD READ/WRITE	22
23	32	/	MEMORY FIELDS ARE UTILIZED, THE FACT THAT A PORTION OF THE SYSTEM DEVICE	23
24	33	/	HANDLER IS ACTUALLY IMPLEMENTED AS A ROM IS UNDETECTABLE TO THE REST OF THE	24
25	34	/	OPERATING SYSTEM; THE ROM HAS NO IMPACT ON CONVENTIONAL MEMORY SIZE. THERE IS	25
26	35	/	NO GENERAL EXPECTATION OF THE AVAILABILITY OF THE HIGHEST TWO PAGES OF MEMORY	26
27	36	/	[77400-77777] UNLESS ALL LOWER ADDRESSES IN THE ENTIRE MEMORY SPACE	27
28	37	/	[00000-77377] EXIST [AND ALL ARE WRITABLE]. AS SUCH, THE ROM IS ESSENTIALLY	28
29	38	/	"INVISIBLE" TO ALL CONFORMING COMPONENTS OF ALL OPERATING SYSTEMS COMPATIBLE	29
30	39	/	WITH THIS UNUSUAL MEMORY CONFIGURATION. THE ONLY PRACTICAL CONSIDERATION IS	30
31	40	/	THE MAXIMUM EFFECTIVE MEMORY SIZE CAN NEVER BE THE FULL 32K ADDRESS SPACE OF	31
32	41	/	THE PDP-8; THE LARGEST CONVENTIONAL MEMORY SIZE POSSIBLE BECOMES 28K.	32
33	42			33
34	43	/	THE MR8E-C WAS DESIGNED FOR USE IN SMALLER CONFIGURATIONS; IF MORE MEMORY IS	34
35	44	/	AVAILABLE, VARIATIONS OF THE SYSTEM CONFIGURATION WILL GENERALLY BE POSSIBLE	35
36	45	/	TO TAKE ADVANTAGE OF ADDITIONAL MEMORY FIELDS. THUS, ON LARGER SYSTEMS, THE	36
37	46	/	MAXIMUM MEMORY SIZE ACHIEVABLE WOULD BE THE USUAL 32K. [NOTE: DURING THE	37
38	47	/	ECONOMIC LIFETIME OF THE TD8E AND COMPATIBLE PDP-8 MODELS, AN ADDITIONAL 4K OF	38
39	48	/	MEMORY MIGHT BE PROHIBITIVELY EXPENSIVE COMPARED TO THE LOWER COST OF THE	39
40	49	/	MR8E-C ROM; THUS, FOR MANY CUSTOMERS WITH MODEST BUDGETS, THE MR8E-C PROVED TO	40
41	50	/	BE A COST-EFFECTIVE ALTERNATIVE.]	41
42	51			42
43	52	/	FOR SOME CUSTOMERS, THE CONVENIENCE OF BOOTING THE SYSTEM WITHOUT HAVING TO	43
44	53	/	TOGGLE IN A BOOTSTRAP ROUTINE JUSTIFIED THE MI8E DIODE-BASED BOOTSTRAP LOADER,	44
45	54	/	A DEVICE WHICH IS DIRECTLY INITIATED FROM THE PDP-8 FRONT PANEL; THE MR8E-C	45
46	55	/	ROM HAS A BUILT-IN BOOTSTRAP THAT IS FUNCTIONALLY EQUIVALENT. THUS, FOR A	46
47	56	/	WHOLE CLASS OF ECONOMY-MINDED CUSTOMERS, THE TD8E WITH MR8E-C CONFIGURATION	47
48	57	/	WAS A WISE CHOICE.	48
49				49
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1	59	/	THERE ARE AT LEAST FOUR OPERATING SYSTEMS THAT SUPPORT THIS CONFIGURATION:	1
2	60			2
3	61	/	1) EDU-SYSTEM 15/30 IS AN OPERATING SYSTEM BASED ON EDU-SYSTEM 30, WHICH	3
4	62	/	IN TURN IS BASED ON POLY BASIC; ALL OF THESE SYSTEMS ONLY SUPPORT THE	4
5	63	/	BASIC COMPUTER LANGUAGE. MOST OF THE WORK ON THESE SYSTEMS WAS DONE	5
6	64	/	BY RICHARD LARY AND LENNY ELEKMAN WHILE STUDENTS AT THE POLYTECHNIC	6
7	65	/	INSTITUTE OF BROOKLYN.	7
8	66			8
9	67	/	ALL OF THESE SYSTEMS ARE DESIGNED TO RUN ON 4K MACHINES; ANY EXTENDED	9
10	68	/	MEMORY FIELDS [IF PRESENT] ARE IGNORED. WHEN THE SYSTEM DEVICE IS THE	10
11	69	/	TD8E SIMPLE DECTAPE CONTROLLER, THE ROM ALLOWS SUPPORT OF THIS	11
12	70	/	SPECIFIC CONFIGURATION. [NOTE: FIELD 7 OF READ/WRITE MEMORY IS NOT	12
13	71	/	ALLOWED SINCE MEMORY LOCATIONS 77400-77777 ARE ALWAYS OCCUPIED BY THE	13
14	72	/	MR8E-C ROM.]	14
15	73			15
16	74	/	NOTE: IN GENERAL, EDU-SYSTEM 15/30 MAY REQUIRE THE KK8F OR KK8A CPU	16
17	75	/	DUE TO THE USE OF INSTRUCTIONS NOT FOUND ON PDP-8 MODELS PRIOR TO THE	17
18	76	/	PDP-8/E; HOWEVER, IT IS STRONGLY RECOMMENDED IN THE CASE OF THE MR8E-C	18
19	77	/	ROM-BASED CONFIGURATION TO ONLY USE THE KK8F CPU. THIS WOULD EXCLUDE	19
20	78	/	THE PDP-8/A 400 SERIES BUT WOULD ALLOW THE PDP-8/A 600 SERIES WHICH	20
21	79	/	USES THE KK8F CPU AS FOUND IN THE PDP-8/E, PDP-8/F AND PDP-8/M. THE	21
22	80	/	SAME RECOMMENDATION APPLIES TO THE OTHER SUPPORTED OPERATING SYSTEMS.	22
23	81	/	THE KK8A CPU AS FOUND IN THE PDP-8/A 400 SERIES MAY NOT BE FAST ENOUGH	23
24	82	/	FOR RELIABLE TD8E OPERATION DUE TO THE SLOWER SPEED OF THE CPU AND THE	24
25	83	/	OVERHEAD ASSOCIATED WITH THE MR8E-C ROM-BASED CODE.	25
26	84			26
27	85	/	2) OS/8 CAN BE CONFIGURED TO RUN ON ANY MODEL PDP-8 WITH AT LEAST 8K OF	27
28	86	/	MEMORY. [12K IS REQUIRED TO RUN OS/8 BATCH.] WHEN THE MR8E-C ROM IS	28
29	87	/	USED WITH THE TD8E, THIS DESIRABLE CONFIGURATION IS ACHIEVABLE. AT	29
30	88	/	LEAST 12K OF MEMORY IS NEEDED IN OS/8 IF THE MR8E-C ROM IS NOT USED	30
31	89	/	[WHEN THE TD8E IS USED AS THE SYSTEM DEVICE] WHICH IMPACTS NEGATIVELY	31
32	90	/	ON CERTAIN OS/8 CONFIGUURATIONS. NOTE: CERTAIN OS/8 CONFIGURATIONS	32
33	91	/	REQUIRE 12K BECAUSE THERE IS NO ANALOG TO THE MR8E-C AVAILABLE. PDP-8	33
34	92	/	PERIPHERALS ARE SOMETIMES INADEQUATELY DESIGNED AND WILL LEAD TO	34
35	93	/	DECREASED OS/8 PERFORMANCE; THE TD8E WITH THE MR8E-C ALLOWS OS/8 TO	35
36	94	/	AVOID THIS ISSUE ON SMALLER MACHINE CONFIGURATIONS.	36
37	95			37
38	96	/	3) COS-300/310 CAN BE CONFIGURED TO RUN ON ANY MODEL PDP-8 WITH AT LEAST	38
39	97	/	8K MEMORY. [INTERNALLY, IT IS LARGELY BASED ON OS/8 AND ALL OF THE	39
40	98	/	CONSIDERATIONS DISCUSSED IN 2) ABOVE APPLY TO COS-300/310 AS WELL.]	40
41	99			41
42	100	/	4) P?S/8 IS AN OPERATING SYSTEM THAT CAN RUN ON ANY MODEL PDP-8 WITH AT	42
43	101	/	LEAST 4K OF RAM MEMORY. VARIOUS P?S/8 COMPONENTS AND SYSTEM PROGRAMS	43
44	102	/	CAN TAKE ADVANTAGE OF ALL AVAILABLE MEMORY UP TO 32K. IN THE SPECIFIC	44
45	103	/	CONFIGURATION OF THE TD8E AND THE MR8E-C ROM, MEMORY SIZE IS LIMITED	45
46	104	/	TO AT MOST 28K DUE TO THE PRESENCE OF THE ROM IN FIELD 7. P?S/8 CAN	46
47	105	/	RUN ADEQUATELY IN 4K AND ALLOWS GENERAL-PURPOSE USE IN ANY SUPPORTED	47
48	106	/	LANGUAGE.	48
49				49
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51				51
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55				55
56				56
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1	108	/	NUMERIC LOAD DEFINITIONS.	1
2	109			2
3	110	7346	NL7775= CLA CLL CMA RTL /LOAD AC WITH 7775.	3
4	111	7240	NL7777= CLA CMA /LOAD AC WITH 7777.	4
5	112			5
6	113	/	MR8E-C ROM DEFINITIONS.	6
7	114			7
8	115	/	THE MR8E-C ROM IS LOCATED AT THE HIGHEST MEMORY LOCATIONS OF THE 32K ADDRESS	8
9	116	/	SPACE; AS SUCH, IT WILL NEVER AFFECT ANY COMPATIBLE OPERATING SYSTEM THAT	9
10	117	/	DYNAMICALLY DETERMINES MEMORY SIZE BY DISCOVERING COMPLETE FIELDS OF	10
11	118	/	READ/WRITE MEMORY.	11
12	119			12
13	120	7552	F2SET= 7552 /CONTINUATION OF FUNCTION CALL ADDRESS.	13
14	121	7340	GOADDR= 7340 /FIELD 0 ADDRESS FOR BOOTSTRAP STORE ROUTINE.	14
15	122	7400	ROMADR= 7400 /ROM LOADS INTO 77400-77777.	15
16	123	7470	ROMBOOT=7470 /ROM-BASED BOOTSTRAP ROUTINE ADDRESS.	16
17	124	0070	ROMFLD= 70 /ROM LOADS INTO FIELD 7.	17
18	125	7405	ROMGO= 7405 /RETRY ADDRESS AFTER ERRORS.	18
19	126			19
20	127	/	SOURCE CONTROL DEFINITIONS.	20
21	128			21
22	129	/	A SKELETAL COPY OF MEMORY LOCATIONS IN FIELD 0 RESERVED BY THE MR8E-C ROM IS	22
23	130	/	NORMALLY CREATED AS PART OF THIS ASSEMBLY. [NOTE: NO ACTUAL BINARY CODE IS	23
24	131	/	GENERATED; THIS IS USED ONLY TO ADD CLARITY TO THE OVERALL PROGRAM LISTING.]	24
25	132	/	THIS ADDITIONAL LISTING SECTION CAN BE DISABLED BELOW; THIS ALLOWS INCLUSION	25
26	133	/	OF THE MR8E-C ROM SOURCE CODE INTO ANOTHER ASSEMBLY FOR THE PURPOSES OF	26
27	134	/	CREATING THE BINARY CODING OF THE SYSTEM DEVICE HANDLER OF ANY OF THE	27
28	135	/	COMPATIBLE OPERATING SYSTEMS.	28
29	136			29
30	137		IFNDEF SYSKEL <SYSKEL=1> /GENERATE THE SYSTEM DEVICE HANDLER SKELETON.	30
31	139	/	SYSKEL= 0 /DO NOT GENERATE THE DEVICE HANDLER SKELETON.	31
32	140			32
33	141	/	NOTE: THE ABILITY TO CONDITIONALLY SPECIFY AN ALTERNATE FIELD FOR THE BINARY	33
34	142	/	OUTPUT HAS BEEN REMOVED TO ENHANCE CLARITY; THE SOURCE CODE HAS BEEN VERIFIED	34
35	143	/	ACCURATE BY COMPARING THE BINARY OUTPUT OBTAINED FROM AN ASSEMBLY OF THIS	35
36	144	/	SOURCE CODE TO A BINARY COPY OBTAINED FROM THE MR8E-C HARDWARE MODULE.	36
37	145			37
38	146	/	TD8E SIMPLE DECTAPE INPUT/OUTPUT DEFINITIONS.	38
39	147			39
40	148	0077	SDCODE= 77 /DEVICE CODE FOR TD8E SIMPLE DECTAPE [PRIMARY].	40
41	149			41
42	150	6771	SDSS= SDCODE^10+6001 /SIMPLE DECTAPE SKIP ON SINGLE LINE FLAG.	42
43	151	6772	SDST= SDCODE^10+6002 /SIMPLE DECTAPE SKIP IF TIME ERROR FLAG SET.	43
44	152	6773	SDSQ= SDCODE^10+6003 /SIMPLE DECTAPE SKIP ON QUAD LINE FLAG.	44
45	153	6774	SDLC= SDCODE^10+6004 /SIMPLE DECTAPE LOAD COMMAND REGISTER.	45
46	154	6775	SDLD= SDCODE^10+6005 /SIMPLE DECTAPE LOAD DATA REGISTER.	46
47	155	6776	SDRC= SDCODE^10+6006 /SIMPLE DECTAPE READ COMMAND REGISTER.	47
48	156	6777	SDRD= SDCODE^10+6007 /SIMPLE DECTAPE READ DATA REGISTER.	48
49				49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
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1	158	/	FIELD 0 DEVICE HANDLER DEFINITIONS.	1
2	159			2
3	160	/	THE FOLLOWING LOCATIONS WITHIN THE LAST PAGE OF FIELD 0 [07600-07777] ARE	3
4	161	/	RESERVED BY THE MR8E-C ROM [AND IN SOME CASES ARE THE HEADERS OF COMPLETELY	4
5	162	/	RESERVED SUBROUTINES]. COMPATIBLE SYSTEMS MUST UTILIZE THESE LOCATIONS	5
6	163	/	CONSISTENTLY WITH THEIR FUNCTION WITHIN THE ROM.	6
7	164			7
8	165	/	NOTE: IF ENABLED, THE SKELETAL VERSION OF THE SYSTEM DEVICE HANDLER CONTAINING	8
9	166	/	ALL RELEVANT ROUTINES WILL ILLUSTRATE THE FULL DEPENDENCIES. IN GENERAL, THE	9
10	167	/	ROM-BASED CODING REQUIRES THESE ROUTINES BE LOCATED IN FIELD 0 AS DEFINED.	10
11	168	/	HOWEVER, THE SPECIFIC OPERATING SYSTEM IS RESPONSIBLE FOR PLACING ALL SUCH	11
12	169	/	ROUTINES INTO FIELD 0 READ/WRITE MEMORY WHEN NEEDED.	12
13	170			13
14	171	/	ALL ADDRESSES NOT SHOWN IN THE SKELETAL VERSION ARE AVAILABLE TO PERFORM	14
15	172	/	SYSTEM-DEPENDENT FUNCTIONS ACCORDING TO THEIR SPECIFIC REQUIREMENTS. SEVERAL	15
16	173	/	LOCATIONS ARE REQUIRED TO ALLOW INDIRECT ACCESS TO ROM-BASED CODING; HOWEVER,	16
17	174	/	THE HANDLERS CAN DESIGNATE WHERE TO PLACE THE RELEVANT POINTERS.	17
18	175			18
19	176	7751	BUFADR= 7751 /READ/WRITE BUFFER ADDRESS.	19
20	177	7753	BUFPTR= 7753 /BUFFER POINTER FOR READ AND STORE ROUTINES.	20
21	178	7674	CIFR= 7674 /INTERNAL RETURN ADDRESS FOR WRITING.	21
22	179	7731	DRVUNIT=7731 /DECTAPE DRIVE UNIT.	22
23	180	7704	EQUFUN= 7704 /EQUIVALENCE CHECKSUM ROUTINE.	23
24	181	7750	EQU TMP= 7750 /TEMPORARY FOR EQUIVALENCE CHECKSUM ROUTINE.	24
25	182	7647	GETWRD= 7647 /GET A WORD FROM THE CALLER'S BUFFER ROUTINE.	25
26	183	7752	PAGCNT= 7752 /TRANSFER PAGE COUNT.	26
27	184	7654	PUTWRD= 7654 /PUT A WORD INTO THE CALLER'S BUFFER ROUTINE.	27
28	185	7676	RDQUAD= 7676 /READ A 12-BIT WORD ROUTINE.	28
29	186	7734	SBLK= 7734 /LATEST DECTAPE BLOCK NUMBER.	29
30	187	7733	SCKSUM= 7733 /RUNNING CHECKSUM WORD.	30
31	188	7661	SERR= 7661 /ERROR HANDLER ADDRESS.	31
32	189	7722	SEXIT= 7722 /ROM EXIT WHEN FINISHED.	32
33	190	7732	SFUNCTI=7732 /FUNCTION WORD.	33
34	191	7735	SUNIT= 7735 /DRIVE UNIT FOR LATEST HANDLER CALL.	34
35	192	7730	TRYCNT= 7730 /RETRY COUNTER.	35
36	193	7754	WORDCT= 7754 /WORD COUNTER FOR LATEST DATA BLOCK.	36
37	194	7724	WRQ= 7724 /WRITE QUAD ROUTINE ROM CONTINUATION ADDRESS.	37
38	195	7671	WRQUAD= 7671 /WRITE A 12-BIT WORD ROUTINE.	38
39				39
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1	197	0007	FIELD	ROMFLD%10	/THE ROM LOADS INTO THIS FIELD.	1
2	198					2
3	199	*7400	*ROMADR		/LOWEST ADDRESS OF THE ROM-BASED CODE.	3
4	200					4
5	201	/		COMES HERE TO SEARCH FOR THE NEXT BLOCK IN A MULTIPLE BLOCK TRANSFER.		5
6	202					6
7	203	077400 1663	NXTBLK, TAD I	PBUFADR/(BUFADDRESS)	/GET THE BUFFER POINTER ADDRESS.	7
8	204	077401 1257	TAD	ROM200/(200)	/UPDATE TO NEXT PAGE.	8
9	205	077402 3663	DCA I	PBUFADR/(BUFADDRESS)	/STORE THE BUFFER POINTER ADDRESS BACK.	9
10	206	077403 2664	ISZ I	PSBLK/(SBLK)	/BUMP TO NEXT BLOCK.	10
11	207	077404 7120	STL		/SECONDARY TRANSFERS ALWAYS SEARCH FORWARD.	11
12	208					12
13	209	/		COMES HERE WITH THE LINK SET TO THE COMPLEMENT OF THE DESIRED DIRECTION.		13
14	210					14
15	211	077405 7232	ROMGO, CLA CML RTR		/INVERT LINK AND MOVE TO DIRECTION BIT.	15
16	212	077406 1312	TAD	ROM1200/(1200)	/ADD ON GO BIT [ONLY HIGHEST FOUR BITS USED].	16
17	213	077407 1665	TAD I	PSUNIT/(SUNIT)	/ADD ON DRIVE UNIT BIT.	17
18	214	077410 6774	SDLC		/LOAD THE COMMAND REGISTER.	18
19	215					19
20	216	/		WE NEED TO DELAY TWO TAPE WORDS BEFORE CHECKING THE MARK REGISTER.		20
21	217					21
22	218	077411 6773	SDSQ		/FLAG UP?	22
23	219	077412 5211	JMP	.-1	/NO, WAIT FOR IT.	23
24	220	077413 6777	SDRD		/YES, CLEAR THE FLAG.	24
25	221	077414 6773	SDSQ		/FLAG UP?	25
26	222	077415 5214	JMP	.-1	/NO, WAIT FOR IT.	26
27	223	077416 6777	SDRD		/YES, CLEAR THE FLAG.	27
28	224					28
29	225	/		CHECK FOR END ZONE OR BLOCK MARK VALUES.		29
30	226					30
31	227	077417 6771	SRCHMOR, SDSS		/SINGLE LINE FLAG UP?	31
32	228	077420 5217	JMP	.-1	/NO, WAIT FOR IT.	32
33	229	077421 6776	SDRC		/YES, READ THE COMMAND REGISTER.	33
34	230	077422 7106	CLL RTL		/MOVE THE DIRECTION BIT TO THE LINK; THE	34
35	231				/MARK WINDOW IS MOVED TO AC BITS[4-9].	35
36	232	077423 0260	AND	ROM374/(374)	/ISOLATE MARK TRACK BITS.	36
37	233	077424 1242	TAD	ROM7670/(-110)	/COMPARE TO END ZONE MARK VALUE.	37
38	234	077425 7450	SNA		/SKIP IF OTHER.	38
39	235	077426 5244	JMP	ENDZONE	/HANDLE END ZONE THERE.	39
40	236	077427 1261	TAD	ROM7760/(-130+110)	/COMPARE TO BLOCK MARK VALUE.	40
41	237	077430 7640	SZA CLA		/SKIP IF IT MATCHES.	41
42	238	077431 5217	JMP	SRCHMORE	/KEEP GOING IF NEITHER MATCH.	42
43						43
44						44
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1	240	/	A BLOCK MARK WAS DETECTED WHILE SEARCHING FOR THE DESIRED BLOCK. THE LATEST	1
2	241	/	BLOCK VALUE AND DIRECTION OF TAPE MOTION WILL BE TESTED ACCORDINGLY.	2
3	242			3
4	243	/	DECTAPE SEARCHING RULES.	4
5	244			5
6	245	/	1) WHILE SEARCHING FORWARDS, IF THE LATEST BLOCK IS LOWER THAN THE	6
7	246	/	DESIRED BLOCK, CONTINUE THE SEARCH MOVING FORWARDS.	7
8	247			8
9	248	/	2) WHILE SEARCHING FORWARDS, IF THE LATEST BLOCK IS THE DESIRED BLOCK,	9
10	249	/	THEN SWITCH TO READ OR WRITE AS REQUIRED.	10
11	250			11
12	251	/	3) WHILE SEARCHING FORWARDS, IF THE LATEST BLOCK IS HIGHER THAN THE	12
13	252	/	DESIRED BLOCK, THE TAPE MUST BE REVERSED. [NOTE: IF THE FORWARD END	13
14	253	/	ZONE IS DETECTED, THE TAPE MOTION MUST BE REVERSED.]	14
15	254			15
16	255	/	4) WHILE SEARCHING IN REVERSE, IF THE LATEST BLOCK IS HIGHER THAN THE	16
17	256	/	DESIRED BLOCK MINUS THE UNDERSHOOT FACTOR, CONTINUE THE SEARCH	17
18	257	/	MOVING IN REVERSE.	18
19	258			19
20	259	/	5) WHILE SEARCHING IN REVERSE, IF THE LATEST BLOCK IS AT THE BLOCK EQUAL	20
21	260	/	TO THE DESIRED BLOCK MINUS THE UNDERSHOOT FACTOR, THE TAPE SHOULD BE	21
22	261	/	TURNED AROUND. ONCE THE TAPE IS MOVING FORWARDS AND UP TO SPEED, 2)	22
23	262	/	ABOVE WILL APPLY. [NOTE: FOR VERY LOW BLOCK NUMBERS, THIS WILL NEVER	23
24	263	/	HAPPEN; HOWEVER, THE END ZONE WILL BE DETECTED AND THE TAPE MOTION	24
25	264	/	MUST BE REVERSED. THERE IS SUFFICIENT INTER-BLOCK ZONE WORDS JUST	25
26	265	/	AFTER THE END ZONE TO ALLOW DETECTION OF BLOCK 0000 AFTER TURNING THE	26
27	266	/	TAPE AROUND IN THE END ZONE.]	27
28	267			28
29	268	/	6) WHILE SEARCHING IN REVERSE, IF THE LATEST BLOCK NUMBER IS LOWER THAN	29
30	269	/	THE DESIRED BLOCK MINUS THE UNDERSHOOT FACTOR, THE TAPE MOTION MUST BE	30
31	270	/	REVERSED. ONCE THE TAPE IS MOVING FORWARDS AND UP TO SPEED, 1) ABOVE	31
32	271	/	APPLIES UNLESS NOW AT THE DESIRED BLOCK, IN WHICH CASE 2) APPLIES.	32
33	272			33
34	273	/	NOTE: IN MOST DECTAPE SEARCH IMPLEMENTATIONS, THE UNDERSHOOT FACTOR IS USUALLY	34
35	274	/	0002; HOWEVER, THE MORE CONSERVATIVE VALUE OF 0003 IS USED IN MOST TD8E	35
36	275	/	APPLICATIONS. IT IS THEORIZED THAT THE EARLIEST TD8E SEARCH ROUTINE WAS	36
37	276	/	COPIED FROM A VERY OLD IMPLEMENTATION OF A DEVICE HANDLER FOR TC01/TC08	37
38	277	/	DECTAPE THAT WAS OBSOLETE; MOST TC01/TC08 SEARCH ROUTINES USE AN UNDERSHOOT	38
39	278	/	FACTOR OF 0002. THERE ARE NO REPORTED PROBLEMS IN ANY SITUATION WHERE	39
40	279	/	PROPERLY ADJUSTED TU55 OR TU56 DRIVES ARE USED. ALL KNOWN TD8E ROUTINES	40
41	280	/	WRITTEN BY DEC PERSONNEL CLEARLY IMITATE THE ORIGINAL TD8E SUBROUTINE THAT WAS	41
42	281	/	RELEASED PRIOR TO THE MR8E-C ROM. AS SUCH, TURN-AROUND PERFORMANCE ON ALL	42
43	282	/	TD8E SYSTEMS RUNNING DEC SOFTWARE IS SOMEWHAT SLUGGISH.	43
44	283			44
45	284	/	SEARCH ALGORITHM DETAILS.	45
46	285			46
47	286	/	THE LATEST BLOCK IS COMPARED TO THE DESIRED BLOCK USING ONE'S COMPLEMENT	47
48	287	/	ARITHMETIC; THE LINK IS CLEAR IF THE CURRENT TAPE MOTION IS FORWARD AND SET IF	48
49	288	/	THE CURRENT TAPE MOTION IS IN THE REVERSE DIRECTION.	49
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57				57
58				58

1	290	/	POSSIBLE SCENARIOS:		1
2	291				2
3	292	/	1)	THE TAPE IS MOVING FORWARDS AND POSITIONED AT A BLOCK LOWER THAN THE	3
4	293	/		DESIRED BLOCK. THE LINK IS CLEAR INDICATING FORWARD MOTION. THE	4
5	294	/		COMPARISON FAILS, AND THE LINK IS SET. AS SUCH, THE SEARCH CONTINUES	5
6	295	/		WITHOUT CHANGING THE DIRECTION OF MOTION.	6
7	296				7
8	297	/	2)	THE TAPE IS MOVING FORWARDS AND POSITIONED AT THE DESIRED BLOCK. THE	8
9	298	/		LINK IS CLEAR INDICATING FORWARD MOTION. THE COMPARISON SUCCEEDS AND	9
10	299	/		THE LINK IS SET. AS SUCH, THE TAPE CAN BE READ TO LOCATE ADDITIONAL	10
11	300	/		DATA WORDS TO POSITION THE TAPE FOR EITHER READING OR WRITING.	11
12	301				12
13	302	/	3)	THE TAPE IS MOVING FORWARDS AND POSITIONED PAST THE DESIRED BLOCK.	13
14	303	/		THE LINK IS CLEAR INDICATING FORWARD MOTION. THE COMPARISON FAILS AND	14
15	304	/		THE LINK STAYS CLEAR. AS SUCH, THE TAPE MOTION NEEDS TO BE REVERSED.	15
16	305				16
17	306	/	4)	THE TAPE IS MOVING IN REVERSE AND POSITIONED AT A BLOCK HIGHER THAN	17
18	307	/		THE DESIRED BLOCK MINUS THE UNDERSHOOT FACTOR. THE LINK IS SET	18
19	308	/		INDICATING REVERSE MOTION. THE COMPARISON FAILS AND THE LINK STAYS	19
20	309	/		SET. AS SUCH, THE SEARCH CONTINUES WITHOUT CHANGING THE DIRECTION OF	20
21	310	/		MOTION.	21
22	311				22
23	312	/	5)	THE TAPE IS MOVING IN REVERSE AND POSITIONED AT THE DESIRED BLOCK	23
24	313	/		MINUS THE UNDERSHOOT FACTOR. THE LINK IS SET INDICATING REVERSE	24
25	314	/		MOTION. THE COMPARISON SUCCEEDS AND THE LINK IS CLEARED. AS SUCH,	25
26	315	/		THE TAPE MOTION NEEDS TO BE REVERSED.	26
27	316				27
28	317	/	6)	THE TAPE IS MOVING IN REVERSE AND POSITIONED AT A BLOCK LOWER THAN THE	28
29	318	/		DESIRED BLOCK MINUS THE UNDERSHOOT FACTOR. THE LINK IS SET INDICATING	29
30	319	/		REVERSE MOTION. THE COMPARISON FAILS AND THE LINK IS CLEARED. AS	30
31	320	/		SUCH, THE TAPE MOTION NEEDS TO BE REVERSED.	31
32	321				32
33	322	/	IN ALL CASES, DETECTION OF THE END ZONE REQUIRES REVERSAL OF TAPE DIRECTION.		33
34	323				34
35	324	077432 6777	SDRD	/GET THE LATEST BLOCK NUMBER	35
36	325	077433 7430	SZL	/SKIP IF GOING FORWARDS.	36
37	326	077434 1262	TAD ROM3/(3)	/ADD ON UNDERSHOOT FACTOR IF MOVING BACKWARDS.	37
38	327	077435 7040	CMA	/INVERT FOR ONE'S COMPLEMENT COMPARISON.	38
39	328	077436 1664	TAD I PSBLK/(SBLK)	/COMPARE TO DESIRED BLOCK.	39
40	329	077437 7040	CMA	/INVERT BACK FOR TEST.	40
41	330	077440 7450	SNA	/SKIP IF NOT THE DESIRED BLOCK.	41
42	331	077441 5252	JMP BLKFND	/JUMP IF CORRECT BLOCK TO TEST FURTHER.	42
43	332	077442 7670	ROM7670,SNA SZL CLA	/SKIP IF THE LINK IS CLEAR; THE SNA IS	43
44	333			/EXTRANEIOUS AND USED TO FORM THE CONSTANT 7670.	44
45	334	077443 5217	JMP SRCHMORE	/KEEP GOING IN THE SAME DIRECTION IF LINK SET.	45
46	335				46
47	336	/	THE TAPE IS MOVING IN A DIRECTION COUNTER TO THE CURRENT INTENDED DIRECTION.		47
48	337	/	IF THE TAPE IS MOVING IN REVERSE OR THE REVERSE END ZONE HAS BEEN DETECTED,		48
49	338	/	THE DIRECTION OF MOTION MUST BE REVERSED. THE LINK IS CLEAR WHICH WILL CAUSE		49
50	339	/	THE TAPE TO BE TURNED AROUND.		50
51	340				51
52	341	/	IF THE TAPE IS MOVING FORWARD OR THE FORWARD END ZONE HAS BEEN DETECTED, AN		52
53	342	/	ERROR CONDITION IS INDICATED [WHICH CAN BE TOLERATED UP TO THREE TIMES].		53
54					54
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1	344	077444	6776	ENDZONE, SDR	/READ THE COMMAND REGISTER.	1
2	345	077445	7106	CLL RTL	/DIRECTION BIT TO LINK.	2
3	346	077446	7630	SZL CLA	/SKIP IF TOO FAR UP THE TAPE OR IN THE END ZONE.	3
4	347	077447	5205	JMP ROMGO	/GO TURN THE TAPE AROUND THERE AND CONTINUE THE	4
5	348				/SEARCH OPERATION.	5
6	349					6
7	350			/ AT THIS POINT, MOVING FORWARD IS CONSIDERED AN ERROR.		7
8	351					8
9	352	077450	6202	CIF 00	/GOING TO FIELD 0.	9
10	353	077451	5666	JMP I PSERR/(SERR)	/CHECK THE RETRY COUNT AND CONTINUE THERE.	10
11	354					11
12	355			/ WE ARE POSITIONED AT THE PROPER BLOCK NUMBER; CHECK IF THE TAPE IS MOVING		12
13	356			/ FORWARD; IF MOVING IN THE REVERSE DIRECTION, THE LINK IS CLEAR WHICH WILL		13
14	357			/ ALLOW THE TAPE TO TURN AROUND.		14
15	358					15
16	359	077452	7630	BLKFND, SZL CLA	/SKIP IF MOVING FORWARD AT THE DESIRED BLOCK.	16
17	360	077453	5205	JMP ROMGO	/GO THERE TO TURN THE TAPE AROUND.	17
18	361					18
19	362			/ THE TAPE IS PROPERLY POSITIONED; CONTINUE SETTING UP THE DATA TRANSFER.		19
20	363					20
21	364	077454	1663	TAD I PBUFADR/(BUFADDRESS)	/GET THE BUFFER ADDRESS FOR THE DATA TRANSFER.	21
22	365	077455	5667	JMP I PRWCONT/(RWCONT)	/CONTINUE THERE.	22
23	366					23
24	367	077456	7600	ROM7600, 7600	/CONSTANT 7600.	24
25	368	077457	0200	ROM200, 200	/CONSTANT 0200.	25
26	369	077460	0374	ROM374, 374	/CONSTANT 0374.	26
27	370	077461	7760	ROM7760, 7760	/CONSTANT 7760.	27
28	371	077462	0003	ROM3, 3	/CONSTANT 0003.	28
29	372	077463	7751	PBUFADR, BUFADR	/POINTER TO BUFFER ADDRESS.	29
30	373	077464	7734	PSBLK, SBLK	/POINTER TO LATEST BLOCK TO TRANSFER.	30
31	374	077465	7735	PSUNIT, SUNIT	/POINTER TO SUNIT.	31
32	375	077466	7661	PSERR, SERR	/POINTER TO SERR.	32
33	376	077467	7600	PRWCONT, RWCONT	/POINTER TO THE REST OF THE READ/WRITE CODE.	33
34						34
35						35
36						36
37						37
38						38
39						39
40						40
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1	378	/	BUILT-IN BOOTSTRAP STARTS HERE. SET THE FRONT-PANEL SWITCHES TO 7470. PRESS	1
2	379	/	EXTENDED ADDRESS LOAD, ADDRESS LOAD, CLEAR AND THEN CONTINUE TO GET HERE.	2
3	380			3
4	381	077470 6201 ROMBOOT,CDF 00	/ENSURE FIELD 0 FOR INDIRECT STORAGE.	4
5	382			5
6	383	/	A SUBROUTINE MUST BE COPIED INTO FIELD 0 TO READ THE DATA WORDS ON THE DECTAPE	6
7	384	/	INTO MEMORY ACCORDING TO THE STANDARD TD8E CONVENTIONS.	7
8	385			8
9	386	077471 1346	TAD STRWRD+1+RLFACOR /GET "DCA I STRPTR".	9
10	387	077472 3741	DCA I PSTRW1/(STRWRD+1) /STORE IT IN FIELD 0 WHERE IT EXECUTES.	10
11	388	077473 1347	TAD STRWRD+2+RLFACOR /GET "CIF 70".	11
12	389	077474 3742	DCA I PSTRW2/(STRWRD+2) /STORE IT IN FIELD 0 WHERE IT EXECUTES.	12
13	390	077475 1350	TAD STRWRD+3+RLFACOR /GET "JMP I STRWRD".	13
14	391	077476 3743	DCA I PSTRW3/(STRWRD+3) /STORE IT IN FIELD 0 WHERE IT EXECUTES.	14
15	392	077477 1351	TAD STRPTR+RLFACOR /GET "BOOTBUFFER".	15
16	393	077500 3744	DCA I PSTRPTR/(STRPTR) /STORE IT IN FIELD 0 WHERE IT IS USED.	16
17	394			17
18	395	/	REWIND THE TAPE UNTIL REVERSE END ZONE MARKS ARE FOUND.	18
19	396			19
20	397	077501 1336	TAD RM3000/(3000) /GET REVERSE DIRECTION AND GO BITS [ON UNIT 0].	20
21	398	077502 6774	SDLC /LOAD THE COMMAND REGISTER TO START REWINDING.	21
22	399	077503 6771 EZLOOP, SDSS	/SINGLE LINE FLAG UP?	22
23	400	077504 5303	JMP .-1 /NO, WAIT FOR IT.	23
24	401	077505 6776	SDRC /YES, READ IN THE COMMAND REGISTER.	24
25	402	077506 0337	AND RM77/(77) /JUST MARK WINDOW BITS.	25
26	403	077507 1334	TAD ROM7756/(-22) /COMPARE TO END ZONE MARK VALUE.	26
27	404	077510 7640	SZA CLA /SKIP IF IT MATCHES.	27
28	405	077511 5303	JMP EZLOOP /KEEP GOING IF NOT.	28
29	406			29
30	407	/	THE END ZONE HAS BEEN FOUND WHILE GOING BACKWARDS. WE NEED TO TURN THE	30
31	408	/	DECTAPE AROUND AND SEARCH FORWARD FOR THE 31 CODE [PART OF THE BLOCK MARK AND	31
32	409	/	THE REVERSE GUARD MARK].	32
33	410			33
34	411	077512 1312 ROM1200,TAD ROM1200/(1200)	/SET UNIT 0 AND GO BITS TO MOVE FORWARD; [NOTE:	34
35	412		/ONLY BITS[0-3] ARE USED.]	35
36	413	077513 6774	SDLC /START THE DECTAPE MOVING FORWARD.	36
37	414	077514 6771 LOOP31, SDSS	/SINGLE LINE FLAG UP?	37
38	415	077515 5314	JMP .-1 /NO, WAIT FOR IT.	38
39	416	077516 6776	SDRC /YES, READ IN THE COMMAND REGISTER.	39
40	417	077517 0337	AND RM77/(77) /JUST MARK WINDOW BITS.	40
41	418	077520 1335	TAD ROM7747/(-31) /COMPARE TO 31 CODE.	41
42	419	077521 7640	SZA CLA /SKIP IF IT MATCHES	42
43	420	077522 5314	JMP LOOP31 /GO BACK IF NOT.	43
44				44
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1	422		/		WE ARE POSITIONED AT THE 31 MARK WINDOW VALUE. START STORING DATA WORDS WHERE	1
2	423		/		THEY BELONG. [NOTE: THE FIRST FOUR ARE BEFORE THE DATA WORDS OF BLOCK 0000.]	2
3	424					3
4	425	077523	6773	BTRDLP, SDSQ	/QUAD FLAG UP?	4
5	426	077524	5323	JMP .-1	/NO, WAIT FOR IT.	5
6	427	077525	6777	SDRD	/YES, READ IN A DATA WORD.	6
7	428	077526	6202	CIF 00	/GOING TO FIELD 0.	7
8	429	077527	4740	JMS I PSTRW1/(STRWRD)	/STORE A DATA WORD.	8
9	430	077530	2744	ISZ I PSTRPTR/(STRPTR)	/BUMP TO NEXT; MORE TO DO?	9
10	431	077531	5323	JMP BTRDLP	/YES, KEEP GOING.	10
11	432	077532	6202	CIF 00	/NO, GOING TO FIELD 0.	11
12	433	077533	5200	JMP STARTIT	/STARTUP THE BOOT CODE FROM THE DECTAPE.	12
13	434					13
14	435	077534	7756	ROM7756, 7756	/CONSTANT 7756.	14
15	436	077535	7747	ROM7747, 7747	/CONSTANT 7747.	15
16	437	077536	3000	RM3000, 3000	/CONSTANT 3000.	16
17	438	077537	0077	RM77, 77	/CONSTANT 0077.	17
18	439	077540	7340	PSTRWRD, STRWRD	/POINTER TO "STRWRD" ROUTINE IN FIELD 0.	18
19	440	077541	7341	PSTRW1, STRWRD+1	/POINTER TO "STRWRD+1" IN FIELD 0.	19
20	441	077542	7342	PSTRW2, STRWRD+2	/POINTER TO "STRWRD+2" IN FIELD 0.	20
21	442	077543	7343	PSTRW3, STRWRD+3	/POINTER TO "STRWRD+3" IN FIELD 0.	21
22	443	077544	7344	PSTRPTR, STRPTR	/POINTER TO "STRPTR" IN FIELD 0.	22
23						23
24						24
25						25
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1	445	7545	RMADDR= .	/WHERE THE FOLLOWING CODE IS LOCATED WITHIN	1
2	446			/THE ROM.	2
3	447				3
4	448		NOPUNCH	/FOOL THE ASSEMBLER.	4
5	449				5
6	450	0000	FIELD 0	/WHERE THIS LOADS.	6
7	451				7
8	452		ENPUNCH	/UN-FOOL THE ASSEMBLER.	8
9	453				9
10	454	*7340	RELOC GOADDR	/WHERE THE FOLLOWING CODE WILL BE	10
11	455			/LOADED IN FIELD 0 WHEN EXECUTED.	11
12	456				12
13	457	0205	RLFACT0=RMADDR- .	/RELOCATION FACTOR FOR MOVED WORDS.	13
14	458				14
15	459	/	SUBROUTINE TO STORE DATA WORDS READ IN FROM THE DECTAPE INTO MEMORY; ALL WORDS		15
16	460	/	WILL BE LOCATED HERE WHEN EXECUTED.		16
17	461				17
18	462	007340* 0000	STRWRD, .-. .	/STORE A WORD ROUTINE.	18
19	463	007341* 3744	DCA I STRPTR	/STORE AC INTO MEMORY	19
20	464	007342* 6272	CIF 70	/GOING TO ROM FIELD.	20
21	465	007343* 5740	JMP I STRWRD	/RETURN TO CALLER.	21
22	466				22
23	467	007344* 7354	STRPTR, BOOTBUFFER	/WHERE THE TAPE DATA IS STORED.	23
24	468				24
25	469	7345	GOEND= .	/END OF RAM-LOADED CODE.	25
26	470				26
27	471	*7552	RELOC	/TURN OFF RELOCATION.	27
28	472				28
29	473	0007	FIELD ROMFLD%10	/RESTORE ROM-LOADING FIELD.	29
30	474				30
31	475	*7552	*GOEND-GOADDR+RMADDR	/RESTORE ROM ADDRESS.	31
32					32
33					33
34					34
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56					56
57					57
58					58

1	477	/	COMES HERE TO CONTINUE THE CALL TO THE SYSTEM DEVICE HANDLER.	1
2	478			2
3	479	/	SEVERAL PARAMETERS ARE SETUP BY THE SYSTEM DEVICE HANDLER:	3
4	480			4
5	481	/	1) THE DRIVE UNIT FOR THE TRANSFER IS PLACED IN BIT[0] OF "SUNIT".	5
6	482			6
7	483	/	2) THE FIELD FOR DATA TRANSFER FROM THE CALLER'S DATA BUFFER IS PLACED IN	7
8	484	/	"GETFLD" IN THE FORM OF "CDF XX" WHERE XX IS THE FIELD OF THE CALLER'S	8
9	485	/	DATA BUFFER. THIS INSTRUCTION IS CONTAINED IN THE "GETWRD" ROUTINE	9
10	486	/	LATER CALLED FROM ROM-BASED ROUTINES FOR WRITING TO THE DECTAPE.	10
11	487			11
12	488	/	3) THE FIELD FOR DATA TRANSFERS TO THE CALLER'S DATA BUFFER IS PLACED IN	12
13	489	/	"PUTFLD" IN THE FORM OF "CDF XX" WHERE XX IS THE FIELD OF THE CALLER'S	13
14	490	/	DATA BUFFER. THIS INSTRUCTION IS CONTAINED IN THE "PUTWRD" ROUTINE	14
15	491	/	LATER CALLED FROM ROM-BASED ROUTINES FOR READING FROM THE DECTAPE.	15
16	492			16
17	493	/	4) "BUFPTR" IS SET TO THE LOWEST ADDRESS OF THE CALLER'S DATA BUFFER.	17
18	494			18
19	495	/	5) THE STARTING BLOCK FOR THE LATEST TRANSFER IS PLACED INTO "SBLK". THE	19
20	496	/	VALUE WILL BE INCREMENTED FOR EACH BLOCK TRANSFERRED. [AS SUCH, IT IS	20
21	497	/	POSSIBLE TO DERIVE THE PROPER INITIAL SEARCH DIRECTION FROM THE	21
22	498	/	PREVIOUS VALUE BY COMPARING TO THE LATEST CALLER'S BLOCK VALUE.]	22
23	499			23
24	500	/	6) THE DATA REGISTER IS SET TO THE FUNCTION WORD AS DEFINED IN OS/8,	24
25	501	/	COS-300/310 AND P?S/8 [WITH AN EXCEPTION NOTED BELOW]:	25
26	502			26
27	503	/	A) THE READ/WRITE FUNCTION IS PASSED IN BIT[0]; 1=WRITE, 0=READ.	27
28	504			28
29	505	/	B) THE PAGE COUNT FOR THE CALL IS CONTAINED IN BITS[1-5]; IF THIS	29
30	506	/	VALUE IS SET TO 00000 THEN TRANSFER AN ENTIRE FIELD [32 PAGES].	30
31	507			31
32	508	/	C) BITS[6-10] ARE IGNORED.	32
33	509			33
34	510	/	D) BIT[11] IS SET TO THE INITIAL SEARCH DIRECTION FOR THE TRANSFER.	34
35	511	/	NOTE: OS/8 ONLY SETS THIS BIT IN THE FEW INSTANCES WITHIN OS/8	35
36	512	/	WHERE THIS INFORMATION IS KNOWN ON A STATIC BASIS. OPERATING	36
37	513	/	SYSTEMS SUCH AS P?S/8 MAY CALCULATE THIS VALUE DYNAMICALLY FOR ALL	37
38	514	/	DEVICE HANDLER CALLS TO PROVIDE NOTICABLY SUPERIOR PERFORMANCE.	38
39	515	/	WITHIN EACH OPERATING SYSTEM THERE IS NO UNIVERSALLY AGREED UPON	39
40	516	/	DEFINITION FOR THIS BIT IN TERMS OF THE CALLER'S FUNCTION WORD.	40
41	517	/	THE INITIAL GUESS AS DEFINED ABOVE ONLY APPLIES TO OS/8 AND	41
42	518	/	COS-300/310. WITHIN P?S/8 THIS BIT IS DEFINED AS THE LOGICAL DRIVE	42
43	519	/	UNIT [0 OR 1], THUS THE NORMAL MEANING OF THE BIT MUST BE MASKED.	43
44	520	/	IF THERE IS SUFFICIENT ADDITIONAL CODE SPACE IN THE HANDLER [WHICH	44
45	521	/	MAY BE ENHANCED BY OTHER OPTIONAL SYSTEM COMPONENTS OF THE LOGICAL	45
46	522	/	CONSOLE OVERLAY], THE DYNAMIC INITIAL SEARCH DIRECTION WILL BE	46
47	523	/	CALCULATED AND INCLUDED IN THE DATA REGISTER. IN EDU-SYSTEM 15/30	47
48	524	/	THIS BIT IS ALWAYS CLEAR; AS SUCH, ALL CALLS START THE DECTAPE IN	48
49	525	/	THE REVERSE DIRECTION.	49
50	526			50
51	527	/	NOTE: WHILE OS/8 SPARSELY USES THE POTENTIALLY FORWARD INITIAL SEARCH	51
52	528	/	DIRECTION, THE DYNAMIC METHOD USED IN P?S/8 MAY BE SUCCESSFUL UP TO AS	52
53	529	/	MUCH AS 40% OF THE TIME DEPENDING ON SPECIFIC SYSTEM USAGE [SUCH AS	53
54	530	/	CAREFUL PLACEMENT OF FILES TO TAKE ADVANTAGE OF FORWARD INITIAL SEARCH	54
55	531	/	SITUATIONS].	55
56				56
57				57
58				58

1	533	077552	1665	F2SET,	TAD I	PSUNIT/(SUNIT)	/GET THE DRIVE UNIT WITHOUT THE GO BIT LOADED	1
2	534	077553	6774		SDLC		/INTO THE COMMAND REGISTER [TAPE IS STOPPED].	2
3	535	077554	7346		NL7775		/SETUP THE	3
4	536	077555	3771		DCA I	PTRYCNT/(TRYCNT)	/RETRY COUNTER.	4
5	537	077556	6776		SDRC		/GET THE UNIT BIT.	5
6	538	077557	3772		DCA I	PDRVUNIT/(DRVUNIT)	/STASH IT THERE.	6
7	539	077560	6777		SDRD		/GET FUNCTION WORD BACK FROM THE DATA REGISTER.	7
8	540	077561	7104		CLL RAL		/MOVE UP AND	8
9	541	077562	0256		AND	ROM7600/(7600)	/ISOLATE PAGE BITS.	9
10	542	077563	3773		DCA I	PPAGCNT/(PAGCNT)	/SAVE AS TRANSFER PAGE COUNTER.	10
11	543	077564	6777		SDRD		/GET FUNCTION WORD BACK FROM THE DATA REGISTER.	11
12	544	077565	3774		DCA I	PSFUNCT/(SFUNCT)	/STASH THE FUNCTION WORD.	12
13	545	077566	6777		SDRD		/GET FUNCTION WORD BACK FROM THE DATA REGISTER.	13
14	546	077567	7010		RAR		/INITIAL SEARCH DIRECTION TO LINK.	14
15	547	077570	5205		JMP	ROMGO	/CONTINUE HERE	15
16	548							16
17	549	077571	7730	PTRYCNT, TRYCNT			/POINTER TO WHERE RETRY COUNT IS STORED.	17
18	550	077572	7731	PDRVUNI, DRVUNIT			/POINTER TO WHERE DRIVE UNIT IS STORED.	18
19	551	077573	7752	PPAGCNT, PAGCNT			/POINTER TO WHERE TRANSFER PAGE COUNT IS STORED.	19
20	552	077574	7732	PSFUNCT, SFUNCT			/POINTER TO WHERE FUNCTION WORD IS STORED.	20
21	553							21
22	554	077575	0000	ZBLOCK	.&7600+200-.		/EMPTY SPACE.	22
23								23
24								24
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58								58

1	556		*7600	PAGE		/NEXT PAGE.	1
2	557						2
3	558		/	COMES HERE TO CONTINUE THE LATEST READ/WRITE OPERATION.			3
4	559						4
5	560		/	THE AC CONTAINS THE LATEST BUFFER ADDRESS.			5
6	561						6
7	562	077600	3754	RWCONT, DCA I	PBUFPtr/(BUFPtr)	/SAVE THE BUFFER ADDRESS.	7
8	563	077601	1223	TAD	RM7600/(-200)	/SETUP THE TRANSFER WORD	8
9	564	077602	3755	DCA I	PWORDCT/(WORDCT)	/COUNT FOR LATEST BLOCK.	9
10	565						10
11	566		/	NOW LOCATE THE REVERSE GUARD MARK.			11
12	567						12
13	568	077603	6771	RGLOOP, SDSS		/FLAG UP?	13
14	569	077604	5203	JMP	.-1	/NO, WAIT FOR IT.	14
15	570	077605	6776	SDRC		/YES, READ THE COMMAND REGISTER.	15
16	571	077606	0241	AND	ROM77/(77)	/JUST THE MARK WINDOW BITS.	16
17	572	077607	1356	TAD	ROMM32/(-32)	/COMPARE TO REVERSE GUARD MARK VALUE.	17
18	573	077610	7640	SZA CLA		/SKIP IF IT MATCHES.	18
19	574	077611	5203	JMP	RGLOOP	/ELSE GO WAIT FOR IT.	19
20	575	077612	1757	TAD I	PSFUN2/(SFUNCTIOn)	/GET THE FUNCTION WORD.	20
21	576	077613	7700	ROM7700, SMA CLA		/SKIP IF WRITING; ALSO USED AS CONSTANT 7700.	21
22	577	077614	5266	JMP	READIT	/JUMP IF READING.	22
23	578	077615	6776	SDRC		/GET THE COMMAND REGISTER.	23
24	579	077616	0307	AND	ROM300/(300)	/JUST WRITE-LOCKOUT AND SELECT/TIME ERROR BITS.	24
25	580	077617	6202	CIF	00	/GOING TO FIELD 0 EITHER WAY.	25
26	581	077620	7640	SZA CLA		/SKIP IF NO ERRORS.	26
27	582	077621	5261	JMP	SERR	/GO THERE TO COMPLAIN ABOUT ERRORS.	27
28	583	077622	4276	JMS	RDQUAD	/READ THE REVERSE GUARD WORD.	28
29	584	077623	7600	RM7600, CLA!400		/CLEAN UP; ALSO USED AS CONSTANT 7600.	29
30	585	077624	1360	TAD	ROM1400/(1400)	/GET GO AND WRITE BITS.	30
31	586	077625	1761	TAD I	PDRVU2/(DRVUNIT)	/ADD ON DRIVE UNIT BIT.	31
32	587	077626	6774	SDLC		/START UP THE WRITE OPERATION.	32
33	588	077627	7240	NL7777		/SET ALL ONES.	33
34	589	077630	6202	CIF	00	/GOING TO FIELD 0.	34
35	590	077631	4271	JMS	WRQUAD	/WRITE REVERSE PCC [CHECKSUM] WORD.	35
36	591	077632	7240	NL7777		/SETUP ALL ONES AGAIN.	36
37	592	077633	3762	DCA I	PSCKSUM/(SCKSUM)	/SETUP THE INITIAL CHECKSUM.	37
38	593	077634	6202	WRILUP, CIF	00	/GOING TO FIELD 0.	38
39	594	077635	4247	JMS	GETWRD	/GET A DATA WORD FROM THE CALLER'S BUFFER.	39
40	595	077636	6203	CIF CDF	00	/GOING TO FIELD 0.	40
41	596	077637	4271	JMS	WRQUAD	/WRITE OUT A DATA WORD.	41
42	597	077640	2754	ISZ I	PBUFPtr/(BUFPtr)	/BUMP TO NEXT.	42
43	598	077641	0077	ROM77, 77		/CONSTANT 0077; PLACED HERE IN CASE IT SKIPS.	43
44	599	077642	2755	ISZ I	PWORDCT/(WORDCT)	/DONE ENOUGH DATA WORDS?	44
45	600	077643	5234	JMP	WRILUP	/NO, GO BACK FOR MORE.	45
46	601	077644	6202	CIF	00	/GOTO FIELD 0.	46
47	602	077645	4271	JMS	WRQUAD	/WRITE 0000 AS 129TH DATA WORD.	47
48	603	077646	5334	JMP	RWCOMMON	/CONTINUE THERE TO WRITE OUT THE CHECKSUM.	48
49							49
50							50
51							51
52							52
53							53
54							54
55							55
56							56
57							57
58							58

1	605		/	COMES HERE WITH CHECKSUM IN AC[0-5] IF WRITING.	1
2	606				2
3	607	077647	6202	WRITEND, CIF 00 /GOING TO FIELD 0.	3
4	608	077650	4271	JMS WRQUAD /WRITE THE PCC WORD.	4
5	609				5
6	610		/	AN EXTRA WRITE IS USED TO ENSURE THE CHECKSUM IS WRITTEN. [APPARENTLY, THE	6
7	611		/	TD8E WRITE OPERATION IS INHERENTLY DOUBLE-BUFFERED, THUS EACH WRITE OPERATION	7
8	612		/	MERELY SETS UP THE DATA FOR THE NEXT WORD.]	8
9	613				9
10	614	077651	6202	CIF 00 /GOING TO FIELD 0.	10
11	615	077652	4271	JMS WRQUAD /THIS FINISHES THE CHECKSUM WRITING.	11
12	616				12
13	617		/	FALLS THROUGH IF WRITING WITH A CLEAR AC. IF READING, THE CHECKSUM IS IN THE	13
14	618		/	AC WHICH [HOPEFULLY] IS ZERO.	14
15	619				15
16	620	077653	6202	READEND, CIF 00 /GOING TO FIELD 0 EITHER WAY.	16
17	621	077654	6772	SDST /TIMING ERROR?	17
18	622	077655	7640	SZA CLA /NOT A TIMING ERROR, IS THE CHECKSUM CLEAR?	18
19	623	077656	5261	JMP SERR /JUMP IF EITHER ERROR.	19
20	624	077657	1763	TAD I PPAGC2/(PAGCNT) /GET THE CURRENT PAGE COUNTER.	20
21	625	077660	1223	TAD RM7600/(-200) /SUBTRACT ONE PAGE.	21
22	626	077661	7450	SNA /SKIP IF STILL MORE TO GO.	22
23	627	077662	5322	JMP SEXIT /ALL ROM OPERATIONS ARE DONE; RETURN TO HANDLER.	23
24	628	077663	3763	DCA I PPAGC2/(PAGCNT) /STORE BACK UPDATED PAGE COUNT.	24
25	629	077664	6272	CIF 70 /STAYING IN FIELD 7 AFTER ALL.	25
26	630	077665	5764	JMP I PNXTBLK/(NXTBLK) /GO DO ANOTHER BLOCK.	26
27	631				27
28	632		/	COMES HERE IF READING.	28
29	633				29
30	634	077666	6202	READIT, CIF 00 /GOING TO FIELD 0.	30
31	635	077667	4276	JMS RDQUAD /READ THE REVERSE GUARD WORD.	31
32	636	077670	6202	CIF 00 /GOING TO FIELD 0.	32
33	637	077671	4276	JMS RDQUAD /READ THE LOCK WORD.	33
34	638	077672	6202	CIF 00 /GOING TO FIELD 0.	34
35	639	077673	4276	JMS RDQUAD /GET THE REVERSE PCC WORD.	35
36	640	077674	0241	AND ROM77/(77) /JUST THE CHECKSUM BITS.	36
37	641	077675	1213	TAD ROM7700/(7700) /ADD ON SET HIGH-ORDER BITS.	37
38	642	077676	3762	DCA I PSCKSUM/(SCKSUM) /STORE INITIAL CHECKSUM VALUE.	38
39	643	077677	6202	READLP, CIF 00 /GOING TO FIELD 0.	39
40	644	077700	4276	JMS RDQUAD /GET A DATA WORD.	40
41	645	077701	6202	CIF 00 /GOING TO FIELD 0.	41
42	646	077702	4304	JMS EQUFUN /INCLUDE DATA IN RUNNING CHECKSUM.	42
43	647	077703	6202	CIF 00 /GOING TO FIELD 0.	43
44	648	077704	4254	JMS PUTWRD /STORE THE DATA WORD IN THE CALLER'S BUFFER.	44
45	649	077705	6201	CDF 00 /BACK TO FIELD 0.	45
46	650	077706	2754	ISZ I PBUFPtr/(BUFPtr) /BUMP TO NEXT ADDRESS.	46
47	651	077707	0300	ROM300, 300 /CONSTANT 0300; PLACED HERE IN CASE IT SKIPS.	47
48	652	077710	2755	ISZ I PWORDCT/(WORDCT) /DONE ALL DATA WORDS?	48
49	653	077711	5277	JMP READLP /NO, KEEP GOING.	49
50					50
51					51
52					52
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54					54
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56					56
57					57
58					58

1	655		/	GET THE 129TH [FINAL] WORD [CHECKSUMMED ONLY, NOT TRANSFERRED].	1
2	656				2
3	657	077712	6202	CIF 00 /GOING TO FIELD 0.	3
4	658	077713	4276	JMS RDQUAD /GET THE FINAL DATA WORD.	4
5	659	077714	6202	CIF 00 /GOING TO FIELD 0.	5
6	660	077715	4304	JMS EQUFUN /INCLUDE FINAL DATA WORD IN RUNNING CHECKSUM.	6
7	661	077716	6202	CIF 00 /GOING TO FIELD 0.	7
8	662	077717	4276	JMS RDQUAD /GET THE PCC WORD.	8
9	663	077720	0213	AND ROM7700/(7700) /JUST CHECKSUM BITS.	9
10	664	077721	6202	CIF 00 /GOING TO FIELD 0.	10
11	665	077722	4304	JMS EQUFUN /INCLUDE IN RUNNING CHECKSUM.	11
12	666	077723	5334	JMP RWCOMMON /CONTINUE THERE.	12
13	667				13
14	668		/	THIS CODE CONTINUES THE WRQUAD SUBROUTINE IN FIELD 0.	14
15	669				15
16	670	077724	6202	WRQ, CIF 00 /GOING TO FIELD 0.	16
17	671	077725	4304	JMS EQUFUN /LATEST DATA WORD .EQU. CHECKSUM.	17
18	672	077726	6773	SDSQ /QUAD FLAG UP?	18
19	673	077727	5326	JMP .-1 /NO, WAIT FOR IT.	19
20	674	077730	6775	SDLD /YES, SEND LATEST DATA WORD.	20
21	675	077731	7300	CLA CLL /CLEAN UP	21
22	676	077732	6202	CIF 00 /GOING TO FIELD 0.	22
23	677	077733	5274	JMP CIFR /CONTINUE THERE TO RETURN TO ORIGINAL CALLER.	23
24	678				24
25	679		/	COMES HERE AT THE END OF THE LATEST BLOCK READ/WRITE OPERATION AFTER THE DATA	25
26	680		/	IS PROCESSED. [NOTE: THE DATA FIELD IS 00.]	26
27	681				27
28	682	077734	7200	RWCOMMO,CLA /CLEAN UP.	28
29	683	077735	1762	TAD I PSCKSUM/(SCHSUM) /GET THE ACCUMULATED CHECKSUM.	29
30	684	077736	7146	CLL CMA RTL;RTL;RTL /INVERT AND MOVE UP SIX BITS.	30
31	685	077737	7006		31
32	686	077740	7006		32
33	687	077741	6202	CIF 00 /GOING TO FIELD 0.	33
34	688	077742	4304	JMS EQUFUN /EQUIVALENCE LOW-ORDER HALF TO HIGH-ORDER HALF.	34
35	689	077743	7300	CLA CLL /CLEAN UP.	35
36	690	077744	1757	TAD I PSFUN2/(SFUNCTION) /GET THE FUNCTION WORD.	36
37	691	077745	7004	RAL /MOVE READ/WRITE BIT TO LINK	37
38	692	077746	7200	CLA /THROW THE REST AWAY.	38
39	693	077747	1762	TAD I PSCKSUM/(SCHSUM) /GET THE FINAL CHECKSUM VALUE.	39
40	694	077750	0213	AND ROM7700/(7700) /JUST THE SIX-BIT CHECKSUM.	40
41	695	077751	7420	SNL /SKIP IF WRITING.	41
42	696	077752	5253	JMP READEND /JUMP IF READING.	42
43	697				43
44	698		/	WE NEED TO WRITE OUT THE PCC MARK (CHECKSUM) WORD.	44
45	699				45
46	700	077753	5247	JMP WRITEND /CONTINUE THERE.	46
47					47
48					48
49					49
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56					56
57					57
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1	702	077754	7753	PBUFPTR, BUFPTR	/POINTER TO DATA BUFFER POINTER.	1
2	703	077755	7754	PWORDCT, WORDCT	/POINTER TO WORD COUNTER.	2
3	704	077756	7746	ROMM32, -32	/CONSTANT 7746.	3
4	705	077757	7732	PSFUN2, SFUNCTION	/POINTER TO FUNCTION WORD.	4
5	706	077760	1400	ROM1400, 1400	/CONSTANT 1400.	5
6	707	077761	7731	PDRVU2, DRVUNIT	/POINTER TO DRIVE UNIT.	6
7	708	077762	7733	PSCKSUM, SCKSUM	/POINTER TO CHECKSUM WORD.	7
8	709	077763	7752	PPAGC2, PAGCNT	/POINTER TO PAGE COUNT.	8
9	710	077764	7400	PNXTBLK, NXTBLK	/POINTER TO NEXT BLOCK ROUTINE.	9
10	711					10
11	712	077765	0000	ZBLOCK .&7600+200- .	/EMPTY SPACE.	11
12	713					12
13	714		0000	ROMEND= .	/END OF ROM CODE.	13
14						14
15						15
16						16
17						17
18						18
19						19
20						20
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56						56
57						57
58						58

1	716				NOPUNCH	/FOOL THE ASSEMBLER.	1
2	717						2
3	718	0000		FIELD	0	/LOADS IN FIELD 0.	3
4	719						4
5	720	/				THE ROM-BASED BUILT-IN BOOTSTRAP IS COMPATIBLE WITH THE STANDALONE TD8E	5
6	721	/				BOOTSTRAP; SOME OF THE LOADING CONVENTIONS ARE SHARED BY BOTH ROUTINES.	6
7	722						7
8	723	/				THE STANDALONE BOOTSTRAP FOLLOWS TO DEFINE THE RELEVANT LOADING AND STARTING	8
9	724	/				ROUTINES.	9
10	725						10
11	726	*7300		*7300		/THIS IS WHERE THE STANDALONE BOOTSTRAP LOADS.	11
12	727						12
13	728	007300*	1312	K1000,	TAD	GET/(2000)	13
14	729					/GET REVERSE DIRECTION BIT. ALSO CONSTANT	14
15	730	007301*	4312		JMS	GET	15
16	731	007302*	4312		JMS	GET	16
17	732	007303*	6773	RD,	SDSQ		17
18	733	007304*	5303		JMP	.-1	18
19	734	007305*	6777		SDRD		19
20	735	007306*	3726		DCA I	WCT	20
21	736	007307*	2326		ISZ	WCT	21
22	737	007310*	5303		JMP	RD	22
23	738					/NO, KEEP GOING.	23
24	739	/				AT THIS POINT BLOCK 0000 HAS BEEN READ INTO 07360-07557 AND BLOCK 0001 HAS	24
25	740	/				BEEN READ INTO 07600-07777. [NOTE: ONLY OS/8 AND COS-300/310 REQUIRE THE	25
26	741	/				BLOCK 0001 DATA LOAD; EDU-SYSTEM 15/30 AND P?S/8 REQUIRE ONLY BLOCK 0000 BE	26
27	742	/				LOADED INTO MEMORY; THE OFFSET WILL BE EXPECTED AND DEALT WITH ACCORDINGLY.]	27
28	743						28
29	744	007311*	5732		JMP I	STRT/(STARTIT)	29
30	745					/GO START IT UP.	30
31	746	/				GET TO DESIRED MARK WINDOW VALUE ROUTINE. [NOTE: INITIALIZED TO SAVE A WORD.]	31
32	747						32
33	748	007312*	2000	GET,	2000	/GET TO DESIRED MARK ROUTINE.	33
34	749	007313*	1300		TAD	K1000/(1000)	34
35	750					/ADD ON UNIT [0] AND GO BITS TO DIRECTION BIT	35
36	751	007314*	6774		SDLC		36
37	752	007315*	6771	MRKSRCH,	SDSS		37
38	753	007316*	5315		JMP	.-1	38
39	754	007317*	6776		SDRC		39
40	755	007320*	0331		AND	K77/(77)	40
41	756					/JUST MARK WINDOW BITS.	41
42	757	/				THE FIRST TIME THIS ROUTINE IS USED, THE REVERSE END ZONE MARK VALUE IS BEING	42
43	758	/				SEARCHED FOR; THE SECOND TIME, THE 31 CODE IS BEING SEARCHED FOR.	43
44	759						44
45	760	007321*	1327	MARKSEL,	TAD	MARK	45
46	761	007322*	7640		SZA	CLA	46
47	762	007323*	5315		JMP	MRKSRCH	47
48	763	007324*	2321		ISZ	MARKSEL	48
49	764	007325*	5712		JMP I	GET	49
50	765					/RETURN TO CALLER.	50
51	766	007326*	7354	WCT,	BOOTBUFFER		51
52	767	007327*	7756	MARK,	-22		52
53	768	007330*	7747		-31		53
54	769	007331*	0077	K77,	77		54
55	770	007332*	7400	STRT,	STARTIT		55
56						/WHERE TO START THE BOOT CODE READ IN.	56
57							57
58							58

1	772	*7354	*7354	/TAPE BLOCK 0000 CONTENTS STARTS LOADING HERE.	1
2	773				2
3	774	007354* 0000	BOOTBUF,ZBLOCK 4	/THESE WORDS ARE LOADED BEFORE THE DATA WORDS.	3
4	775				4
5	776	7360	BLKDATA=.	/THE DATA WORDS IN BLOCK 0000 ARE LOADED HERE.	5
6	777				6
7	778	007360* 0000	ZBLOCK 7400-.	/THE FIRST 0020 WORDS OF BLOCK 0000 LOAD HERE.	7
8	779				8
9	780	7400	STARTIT=.	/THE REST OF THE WORDS OF BLOCK 0000 LOAD HERE	9
10	781			/AND CONTROL IS TRANSFERRED HERE; THE CODE	10
11	782			/LOADED HERE NEEDS TO BE AWARE OF THIS TO	11
12	783			/CONTINUE THE SYSTEM-SPECIFIC BOOTUP PROCESS.	12
13					13
14					14
15					15
16					16
17					17
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58					58

1	791	0000	FIELD 0	/WHERE SUPPORT ROUTINES MUST BE LOADED.	1
2	792				2
3	793	*7600	PAGE 37	/IN VARIOUS LOCATIONS OF THE LAST PAGE.	3
4	794				4
5	795	/	CERTAIN ROUTINES WITHIN THE SYSTEM DEVICE HANDLER MUST BE LOCATED AT SPECIFIC		5
6	796	/	ADDRESSES AS DETERMINED WITHIN THE ROM. ALL MEMORY LOCATIONS DEFINED BELOW IN		6
7	797	/	FIELD 0 MUST BE LOADED ACCORDINGLY; IT IS THE RESPONSIBILITY OF THE OPERATING		7
8	798	/	SYSTEM TO LOAD THE CORRECT VALUES INTO THE STATED MEMORY LOCATIONS.		8
9	799				9
10	800	/	THE OPERATING SYSTEM SETS UP THE ROM-DESIGNATED VARIABLES FROM THE PARAMETERS		10
11	801	/	PASSED TO IT ACCORDING TO THE SYSTEM'S SUBROUTINE CALL CONVENTIONS. THIS		11
12	802	/	INCLUDES PASSING SOME ASPECT OF THE FUNCTION WORD OR OTHER DATA IN THE TD8E		12
13	803	/	DATA REGISTER [USING THE "SDLD" INSTRUCTION]. WHEN ALL VARIABLES ARE SET, THE		13
14	804	/	ROM IS ENTERED BY THE FOLLOWING:		14
15	805				15
16	806	/	CDF 00	/SETUP OUR FIELD	16
17	807	/	CIF ROMFLD	/GOING TO ROM FIELD.	17
18	808	/	JMP I PF2SET/(F2SET)	/START UP THE ROM TO INITIATE THE CALL.	18
19	809				19
20	810	/	SUPPORT ROUTINES FOR THE ROM-BASED CODE.		20
21	811				21
22	812	*7647	*GETWRD	/WHERE THIS BELONGS.	22
23	813				23
24	814	007647* 0000	GETWRD, ..	/GET A WORD FROM THE CALLER'S BUFFER ROUTINE.	24
25	815	007650* 6201	GETFLD, ..^10+CDF	/WILL BE "CDF XX" WHERE XX IS THE	25
26	816			/DATA FIELD OF THE CALLER'S BUFFER.	26
27	817	007651* 1753	TAD I BUFPTR	/GET A WORD FROM THE BUFFER.	27
28	818	007652* 6272	CIF ROMFLD	/GOING TO ROM FIELD.	28
29	819	007653* 5647	JMP I GETWRD	/RETURN TO CALLER.	29
30	820				30
31	821	007654* 0000	PUTWRD, ..	/PUT A WORD INTO THE CALLER'S BUFFER ROUTINE.	31
32	822	007655* 6201	PUTFLD, ..^10+CDF	/WILL BE "CDF XX" WHERE XX IS THE	32
33	823			/DATA FIELD OF THE CALLER'S BUFFER.	33
34	824	007656* 3753	DCA I BUFPTR	/PUT A WORD INTO THE BUFFER.	34
35	825	007657* 6272	CIF ROMFLD	/GOING TO ROM FIELD.	35
36	826	007660* 5654	JMP I PUTWRD	/RETURN TO CALLER.	36
37	827				37
38	828	/	ERRORS COME HERE. [NOTE: NOT ALL OPERATING SYSTEMS HANDLE ERRORS IN THE SAME		38
39	829	/	EXACT MANNER; WHAT FOLLOWS IS TYPICAL FOR SOME SUPPORTED SYSTEMS.]		39
40	830				40
41	831	7661	SERR= .	/ERROR HANDLING ROUTINE HERE.	41
42	832				42
43	833	/	CLA CLL	/CLEAN UP.	43
44	834	/	CIF ROMFLD	/GOING TO ROM FIELD.	44
45	835	/	ISZ TRYCNT	/TOO MANY ERRORS?	45
46	836	/	JMP I PROMGO/(ROMGO)	/NO, TRY AGAIN.	46
47	837				47
48	838	/	TOO MANY RETRIES; HANDLE FATAL ERROR HERE.		48
49	839				49
50	840	/	TAD SUNIT	/GET THE CURRENT TAPE UNIT.	50
51	841	/	SDLC	/STOP THE TAPE.	51
52					52
53					53
54					54
55					55
56					56
57					57
58					58

1	843	*7671	*WRQUAD	/WHERE THIS BELONGS.	1
2	844				2
3	845	007671* 0000	WRQUAD, .-. .	/WRITE A 12-BIT WORD TO THE DECTAPE ROUTINE.	3
4	846				4
5	847	/	THE ACTUAL CODE IS IN THE ROM; THIS IS USED ONLY TO ESTABLISH THE PROPER		5
6	848	/	SUBROUTINE LINKAGE AS IT IS CALLED FROM SEVERAL PLACES.		6
7	849				7
8	850	007672* 6272	CIF ROMFLD	/GOING TO THE ROM FIELD.	8
9	851	007673* 5324	JMP WRQ	/CONTINUE THERE.	9
10	852				10
11	853	/	COMES BACK HERE FROM THE ROM-BASED CODE THAT CARRIES OUT THE SUBROUTINE CODE.		11
12	854				12
13	855	007674* 6272	CIFR, CIF ROMFLD	/GOING TO THE ROM FIELD.	13
14	856	007675* 5671	JMP I WRQUAD	/RETURN TO CALLER.	14
15	857				15
16	858	007676* 0000	RDQUAD, .-. .	/READ A 12-BIT WORD FROM THE DECTAPE ROUTINE.	16
17	859	007677* 6773	SDSQ	/QUAD LINE FLAG UP?	17
18	860	007700* 5277	JMP .-1	/NO, WAIT FOR IT.	18
19	861	007701* 6777	SDRD	/YES, READ IN THE LATEST 12-BIT WORD.	19
20	862	007702* 6272	CIF ROMFLD	/GOING TO ROM FIELD.	20
21	863	007703* 5676	JMP I RDQUAD	/RETURN TO CALLER.	21
22	864				22
23	865	007704* 0000	EQUFUN, .-. .	/EQUIVALENCE DATA WITH CHECKSUM ROUTINE.	23
24	866	007705* 7040	CMA	/INVERT THE DATA AND ,XOR. THE DATA WORD.	24
25	867	007706* 3350	DCA EQU TMP	/SAVE IT.	25
26	868	007707* 1350	TAD EQU TMP	/GET IT BACK.	26
27	869	007710* 0333	AND SCKSUM	/GET THE CARRY BITS.	27
28	870	007711* 7041	CIA	/INVERT FOR SUBTRACTION.	28
29	871	007712* 7104	CLL RAL	/MOVE OVER.	29
30	872	007713* 1350	TAD EQU TMP	/NOW ADD THE LATEST.	30
31	873	007714* 1333	TAD SCKSUM	/ADD THE NON-CARRY BITS TO RUNNING SUM.	31
32	874	007715* 3333	DCA SCKSUM	/STORE BACK.	32
33	875	007716* 1350	TAD EQU TMP	/GET THE LATEST DATA BACK.	33
34	876	007717* 7140	CLL CMA	/NOW HAVE CLEAR LINK AND THE LATEST WORD.	34
35	877	007720* 6272	CIF ROMFLD	/GOING TO ROM FIELD.	35
36	878	007721* 5704	JMP I EQUFUN	/RETURN TO CALLER.	36
37	879				37
38	880	/	THE ROM RETURNS HERE WHEN ALL TAPE OPERATIONS HAVE FINISHED.		38
39	881				39
40	882	7722	SEXIT= .	/SUCCESSFUL CALL COMPLETED.	40
41	883				41
42	884	/	TAD SUNIT	/GET CURRENT TAPE DRIVE UNIT.	42
43	885	/	SDLC	/STOP THE TAPE.	43
44	886				44
45	887	/	OTHER SYSTEM-SPECIFIC OPERATIONS ARE PERFORMED HERE FOLLOWED BY THE RETURN		45
46	888	/	TO THE ORIGINAL CALLER.		46
47					47
48					48
49					49
50					50
51					51
52					52
53					53
54					54
55					55
56					56
57					57
58					58

1	890	/	THE FOLLOWING LOCATIONS MUST BE DEFINED AS REQUIRED BY THE MR8E-C ROM.	1
2	891			2
3	892	*7730	*TRYCNT /WHERE THIS SET OF MEMORY LOCATIONS BELONGS.	3
4	893			4
5	894	007730* 0000	TRYCNT, ... /RETRY COUNTER.	5
6	895	007731* 0000	DRVUNIT, ... /DECTAPE DRIVE UNIT.	6
7	896	007732* 0000	SFUNCTI, ... /FUNCTION WORD BIT[0]; CLEAR=READ, SET=WRITE.	7
8	897		/[NOTE: ALL OTHER BITS ARE IGNORED.]	8
9	898	007733* 0000	SCKSUM, ... /RUNNING CHECKSUM WORD.	9
10	899	007734* 0000	SBLK, ... /LATEST DECTAPE BLOCK.	10
11	900	007735* 0000	SUNIT, ... /DRIVE UNIT FOR LATEST FUNCTION.	11
12	901			12
13	902	*7750	*EQU TMP /WHERE THIS SET OF MEMORY LOCATIONS BELONGS.	13
14	903			14
15	904	007750* 0000	EQU TMP, ... /TEMPORARY FOR EQUIVALENCE ROUTINE.	15
16	905	007751* 0000	BUFADRE, ... /READ/WRITE BUFFER POINTER.	16
17	906	007752* 0000	PAGCNT, ... /TRANSFER PAGE COUNT.	17
18	907	007753* 0000	BUFPTR, ... /BUFFER POINTER FOR WORD TRANSFERS.	18
19	908	007754* 0000	WORDCT, ... /WORD COUNTER FOR LATEST BLOCK TRANSFER.	19
20	909			20
21	910	/	END OF SYSTEM DEVICE HANDLER SKELETON.	21
22	914			22
23	915	\$	/THAT'S ALL FOLK!	23
24				24
25				25
26				26
27				27
28				28
29				29
30				30
31				31
32				32
33				33
34				34
35				35
36				36
37				37
38				38
39				39
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52				52
53				53
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1				1
2				2
3	BLKDAT=7360	READEN 7653	WRQ 7724	3
4	BLKFND 7452	READIT 7666	WRQUAD 7671	4
5	BOOTBU 7354	READLP 7677		5
6	BTRDLP 7523	RGLLOOP 7603		6
7	BUFADR 7751	RLFACT=0205		7
8	BUFPTR 7753	RMADDR=7545		8
9	CIFR 7674	RM3000 7536		9
10	DRVUNI 7731	RM7600 7623		10
11	ENDZON 7444	RM77 7537		11
12	EQUFUN 7704	ROMADR=7400		12
13	EQUTMP 7750	ROMB00 7470		13
14	EZLOOP 7503	ROMEND=0000		14
15	F2SET 7552	ROMFLD=0070		15
16	GET 7312	ROMG0 7405		16
17	GETFLD 7650	ROMM32 7756		17
18	GETWRD 7647	ROM120 7512		18
19	GOADDR=7340	ROM140 7760		19
20	GOEND =7345	ROM200 7457		20
21	K1000 7300	ROM3 7462		21
22	K77 7331	ROM300 7707		22
23	LOOP31 7514	ROM374 7460		23
24	MARK 7327	ROM760 7456		24
25	MARKSE 7321	ROM767 7442		25
26	MRKSR 7315	ROM77 7641		26
27	NL7775=7346	ROM770 7613		27
28	NL7777=7240	ROM774 7535		28
29	NXTBLK 7400	ROM775 7534		29
30	OFF =0001	ROM776 7461		30
31	ON =0000	RWCOMM 7734		31
32	PAGCNT 7752	RWCONT 7600		32
33	PBUFAD 7463	SBLK 7734		33
34	PBUFPT 7754	SCKSUM 7733		34
35	PDRVUN 7572	SDCODE=0077		35
36	PDRVU2 7761	SDLC =6774		36
37	PNXTBL 7764	SDLD =6775		37
38	PPAGCN 7573	SDRC =6776		38
39	PPAGC2 7763	SDRD =6777		39
40	PRWCON 7467	SDSQ =6773		40
41	PSBLK 7464	SDSS =6771		41
42	PSCKSU 7762	SDST =6772		42
43	PSERR 7466	SERR =7661		43
44	PSFUNC 7574	SEXIT =7722		44
45	PSFUN2 7757	SFUNCT 7732		45
46	PSTRPT 7544	SRCHMO 7417		46
47	PSTRWR 7540	STARTI=7400		47
48	PSTRW1 7541	STRPTR 7344		48
49	PSTRW2 7542	STRT 7332		49
50	PSTRW3 7543	STRWRD 7340		50
51	PSUNIT 7465	SUNIT 7735		51
52	PTRYCN 7571	SYSKEL=0001		52
53	PUTFLD 7655	TRYCNT 7730		53
54	PUTWRD 7654	WCT 7326		54
55	PWORDC 7755	WORDCT 7754		55
56	RD 7303	WRILUP 7634		56
57	RDQUAD 7676	WRITEN 7647		57
58				58

1		1
2		2
3	NO ERRORS DETECTED	3
4		4
5	NO LINKS GENERATED	5
6		6
7	5K MEMORY UTILIZED	7
8		8
9	1 FILE CREATED	9
10		10
11	112 SYMBOLS	11
12		12
13		13
14		14
15		15
16		16
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2		2
3		3
4		4
5	70000	5
6	70100	6
7		7
8	70200	8
9	70300	9
10		10
11	70400	11
12	70500	12
13		13
14	70600	14
15	70700	15
16		16
17	71000	17
18	71100	18
19		19
20	71200	20
21	71300	21
22		22
23	71400	23
24	71500	24
25		25
26	71600	26
27	71700	27
28		28
29	72000	29
30	72100	30
31		31
32	72200	32
33	72300	33
34		34
35	72400	35
36	72500	36
37		37
38	72600	38
39	72700	39
40		40
41	73000	41
42	73100	42
43		43
44	73200	44
45	73300	45
46		46
47	73400	47
48	73500	48
49		49
50	73600	50
51	73700	51
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3		3
4		4
5	74000	5
6	74100	6
7		7
8	74200	8
9	74300	9
10		10
11	74400	11
12	74500	12
13		13
14	74600	14
15	74700	15
16		16
17	75000	17
18	75100	18
19		19
20	75200	20
21	75300	21
22		22
23	75400	23
24	75500	24
25		25
26	75600	26
27	75700	27
28		28
29	76000	29
30	76100	30
31		31
32	76200	32
33	76300	33
34		34
35	76400	35
36	76500	36
37		37
38	76600	38
39	76700	39
40		40
41	77000	41
42	77100	42
43		43
44	77200	44
45	77300	45
46		46
47	77400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111	47
48	77500 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111	48
49		49
50	77600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111	50
51	77700 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111	51
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